

假设D2、D4和C1这三块丢失，在编码公式的结果中去掉这三块，得新的公式

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ g_{21} & g_{22} & g_{23} & g_{24} \\ g_{31} & g_{32} & g_{33} & g_{34} \end{bmatrix} \times_{GF(2^m)} \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix} = \begin{bmatrix} D_1 \\ D_3 \\ C_2 \\ C_3 \end{bmatrix}$$

求逆，得到解码公式，恢复丢失的数据块D2和D4

$$\left( \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ g_{21} & g_{22} & g_{23} & g_{24} \\ g_{31} & g_{32} & g_{33} & g_{34} \end{bmatrix} \right)^{-1} \times_{GF(2^m)} \begin{bmatrix} D_1 \\ D_3 \\ C_2 \\ C_3 \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix}$$

利用所有数据块，重新编码丢失的校验块C1

$$\begin{bmatrix} g_{21} & g_{22} & g_{23} & g_{24} \end{bmatrix} \times_{GF(2^m)} \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix} = C_1$$